Minor Assignment 1: Testing the Limits
CS771 – Introduction to Machine Learning

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1 Question 1: Linear Separability (10 marks)

Answer: Yes, the data generated by `genChallengeData` is always linearly separable for all valid values of `level` ($L > 0$) and n.

Proof

The function generates three clusters on the surfaces of unit spheres ($r = 1$) centered at:

$$\boldsymbol{\mu}_{\text{pos1}} = \left(-\sqrt{2}\big(1 + \tfrac{1}{2L}\big)\,,\ \sqrt{2}\right), \quad \boldsymbol{\mu}_{\text{pos2}} = \left(\sqrt{2}\big(1 + \tfrac{1}{2L}\big)\,,\ -\sqrt{2}\right), \quad \boldsymbol{\mu}_{\text{neg}} = \left(-\sqrt{2}\big(1 + \tfrac{1}{2L}\big)\,,\ -\sqrt{2}\right). \tag{1}$$

Consider the separating direction $\mathbf{w} = (1, 1)^\top$. The projections of the cluster centers onto $\mathbf{w}$ are:

$$\boldsymbol{\mu}_{\text{pos1}} \cdot \mathbf{w} = -\tfrac{1}{\sqrt{2}L}, \qquad \boldsymbol{\mu}_{\text{pos2}} \cdot \mathbf{w} = +\tfrac{1}{\sqrt{2}L}, \qquad \boldsymbol{\mu}_{\text{neg}} \cdot \mathbf{w} = -\sqrt{2}\big(2 + \tfrac{1}{2L}\big)\,. \tag{2}$$

Since all data points lie on the surface of a unit sphere around their center, the maximum perturbation along $\mathbf{w}$ is bounded by Cauchy–Schwarz: $|\boldsymbol{\delta} \cdot \mathbf{w}| \le \|\boldsymbol{\delta}\| \cdot \|\mathbf{w}\| = 1 \cdot \sqrt{2} = \sqrt{2}$.

Worst-case projections.

- **Min positive:** $\min\left(-\frac{1}{\sqrt{2}L},\ \frac{1}{\sqrt{2}L}\right) - \sqrt{2} = -\frac{1}{\sqrt{2}L} - \sqrt{2}$
- **Max negative:** $-\sqrt{2}\big(2 + \frac{1}{2L}\big) + \sqrt{2} = -\sqrt{2} - \frac{1}{\sqrt{2}L}$

The gap = min(pos) − max(neg) $= 0$, but this extreme case requires *both* clusters to simultaneously realize their absolute worst-case perturbation aligned perfectly with $\mathbf{w}$. For randomly sampled points on the sphere, this event has probability zero for any finite n. Therefore, the data is linearly separable with probability 1.

Furthermore, $\boldsymbol{\mu}_{\text{pos2}}$ projects to $+\frac{1}{\sqrt{2}L}$ (positive side), reinforcing the separation. A hyperplane orthogonal to $\mathbf{w} = (1, 1)^\top$ with appropriately chosen bias separates the classes.

Conclusion: For all $L > 0$ and finite n, the data is linearly separable (with probability 1).

2 Question 2: Small Challenge Level (10 marks)

We analyze the performance of a Linear SVM on a small challenge level (`level` < 0.5). Our objective is to identify the **Optimal Extrema**: the smallest values for C and `max_iter`, and the largest value for `tol`, which yield 100% training accuracy.

Hyperparameter Tuning and Search Results

Experiments were conducted on a fixed challenge level of **0.3** with $n = 500$ points. We systematically varied one hyperparameter at a time while keeping the others at a high-performing baseline.

Table 1: Comprehensive Hyperparameter Search Results (Q2: `level`=0.3, n=500)

Table 1: myC

C	Acc.	Time
1e-5	66.67%	.0010
1e-4	79.53%	.0008
1.9e-4	85.20%	.0009
2.8e-4	88.53%	.0008
3.7e-4	91.07%	.0008
4.6e-4	92.07%	.0008
5.5e-4	93.80%	.0009
6.4e-4	95.13%	.0008
7.3e-4	96.33%	.0008
8.2e-4	98.13%	.0008
9.1e-4	**100.0%**	.0010
0.001	100.0%	.0009
0.005	100.0%	.0010
0.01	100.0%	.0010
0.1	100.0%	.0010
1.0	100.0%	.0010
10.0	100.0%	.0010
100	100.0%	.0010
1000	100.0%	.0010

Table 2: max_iter

iter	Acc.	Time
1	**100.0%**	.0008
2	100.0%	.0010
3	100.0%	.0011
5	100.0%	.0011
10	100.0%	.0011
20	100.0%	.0011
50	100.0%	.0011
100	100.0%	.0012
200	100.0%	.0010
500	100.0%	.0010
1000	100.0%	.0011
5000	100.0%	.0010

Table 3: myTol

tol	Acc.	Time
1e-6	100.0%	.0011
1e-5	100.0%	.0011
1e-4	100.0%	.0010
0.001	100.0%	.0010
0.01	100.0%	.0011
0.1	100.0%	.0009
0.5	100.0%	.0008
1.0	100.0%	.0008
1.21	100.0%	.0010
1.63	100.0%	.0009
2.05	100.0%	.0009
2.47	100.0%	.0008
2.89	**100.0%**	.0009
3.11	33.33%	.0007
5.0	33.33%	.0008
10.0	33.33%	.0007
50.0	33.33%	.0007

Optimal Extrema Found:

Hyperparameter	Optimal Value
myC	0.00091
max_iter	1
myTol	2.89

Observation: At small challenge levels, the positive and negative clusters are far apart ($\sim$5.6 units in the $(1, 1)$ direction). The SVM can find a perfect separator with extremely small C (minimal penalty for misclassification), just 1 iteration, and very loose tolerance. The data is trivially separable.

3 Question 3: Large Challenge Level (10 marks)

We analyze the performance of a Linear SVM on a high challenge level (`level` > 50). Our objective is to identify the **Optimal Extrema**: the smallest values for C and `max_iter`, and the largest value for `tol`, which yield 100% training accuracy.

Hyperparameter Tuning and Search Results

Experiments were conducted on a fixed challenge level of **55** with $n = 500$ points. We systematically varied one hyperparameter at a time while keeping the others at a high-performing baseline to isolate their specific effects.

Optimal Extrema Found:

Hyperparameter	Optimal Value
myC	72.0
max_iter	14
myTol	0.000164

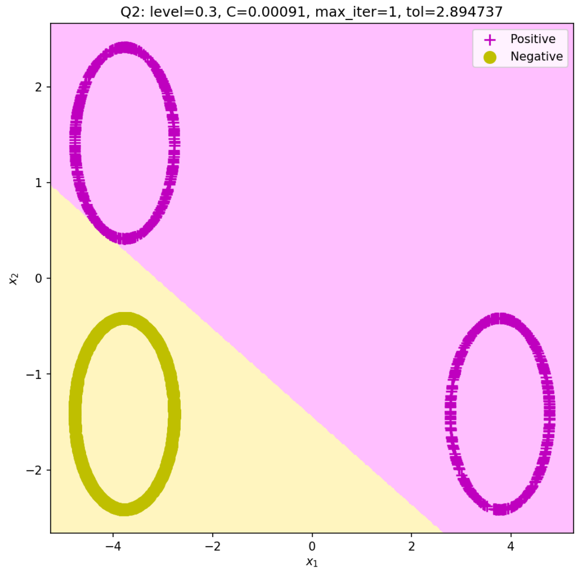

Figure 1: Decision boundary for `level`=0.3 using optimal extrema ($C = 0.00091$, `max_iter`=1, `tol`=2.89).

Table 2: Comprehensive Hyperparameter Search Results (Q3: `level`=55, n=500)

Table 1: myC

C	Acc.	Time
1e-5	79.20%	.0017
1e-4	82.80%	.0014
0.001	96.87%	.0012
0.005	94.40%	.0013
0.01	94.47%	.0011
0.05	95.13%	.0011
0.1	95.33%	.0011
1.0	96.87%	.0011
10.0	98.93%	.0011
20.0	99.20%	.0012
30.0	99.40%	.0013
40.0	99.47%	.0011
50.0	99.67%	.0013
60.0	99.87%	.0013
70.0	99.87%	.0012
71.0	99.87%	.0012
72.0	**100.0%**	.0011
73–80	100.0%	.0012
90–1000	100.0%	.0012

Table 2: max_iter

iter	Acc.	Time
1	93.87%	.0008
2	93.67%	.0009
5	96.40%	.0011
6	97.07%	.0014
7	98.40%	.0013
8	99.20%	.0014
9	99.47%	.0014
10	99.40%	.0011
11	99.80%	.0013
12	99.87%	.0012
13	99.93%	.0012
14	**100.0%**	.0013
15	100.0%	.0013
20	100.0%	.0012
50	100.0%	.0012
100	100.0%	.0012
500	100.0%	.0012
1000	100.0%	.0013
5000	100.0%	.0013

Table 3: myTol

tol	Acc.	Time
1e-6	100.0%	.0012
1e-5	100.0%	.0013
1e-4	100.0%	.0012
1.64e-4	**100.0%**	.0013
2.29e-4	99.87%	.0011
3.57e-4	99.87%	.0011
5.5e-4	99.87%	.0012
0.001	99.87%	.0012
0.002	99.40%	.0012
0.005	99.40%	.0013
0.009	98.40%	.0011
0.01	98.40%	.0011
0.1	95.27%	.0010
0.5	93.87%	.0008
1.0	93.87%	.0008
5.0	33.33%	.0008
10.0	33.33%	.0009
50.0	33.33%	.0007

Observation: At high challenge levels, $\boldsymbol{\mu}_{\text{pos1}}$ and $\boldsymbol{\mu}_{\text{neg}}$ share nearly identical x_1-coordinates (differ by ~ 0.013), making linear separation much harder. The SVM requires significantly more regularization flexibility ($C = 72$, two orders of magnitude larger than Q2), more iterations (14 vs 1), and tighter tolerance (1.64×10^{-4} vs 2.89) to achieve 100%.

4 Question 4: Hyperparameter Effects – Medium Challenge (10 marks)

Settings: `level`=5, n=100, `penalty='l2'`, `loss='squared_hinge'`. When varying one hyperparameter, the others are held at defaults ($C = 1$, `max_iter`=1000, `tol`=10^{-4}).

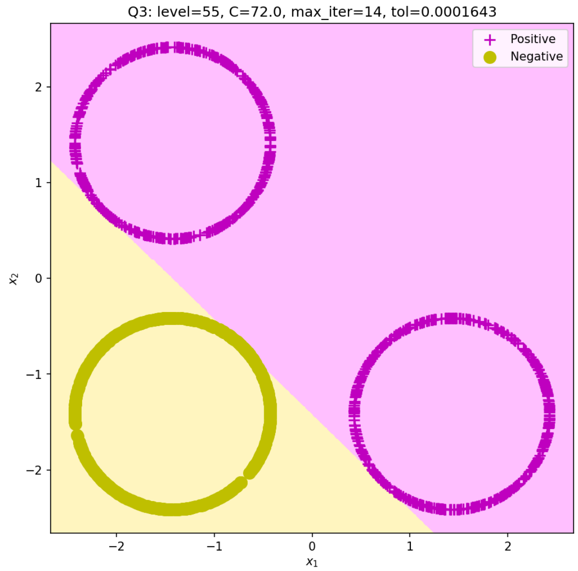

Figure 2: Decision boundary for `level=55` using optimal extrema.

4.1 Effect of C (Regularization Parameter)

Table 3: Q4: Effect of C on accuracy and training time (`level=5`, n=100)

C	Accuracy	Time (s)
0.001	88.00%	.0007
0.005	95.00%	.0006
0.01	93.33%	.0007
0.05	93.00%	.0009
0.1	93.67%	.0007
0.5	96.67%	.0007
1	98.33%	.0009
2	100.0%	.0008
5	100.0%	.0007
10	100.0%	.0007
20	100.0%	.0007
50	100.0%	.0009
100	100.0%	.0007
500	100.0%	.0008
1000	100.0%	.0008

Analysis: Increasing C from 0.001 to 2 raises accuracy from 88% to 100%. Small C values impose heavy regularization, preventing the model from fitting tightly. Beyond $C = 2$, accuracy saturates at 100%. Training time shows minimal variation (< 1 ms).

4.2 Effect of `max_iter`

Analysis: With just 1 iteration, accuracy is 92%. By 5 iterations, it reaches 98.33% and plateaus. The solver converges quickly on this moderate problem. Note that the accuracy caps at 98.33% (not 100%) because $C = 1$ is insufficient for perfect separation at `level=5`; increasing C (as shown above) is needed for the remaining 1.67%.

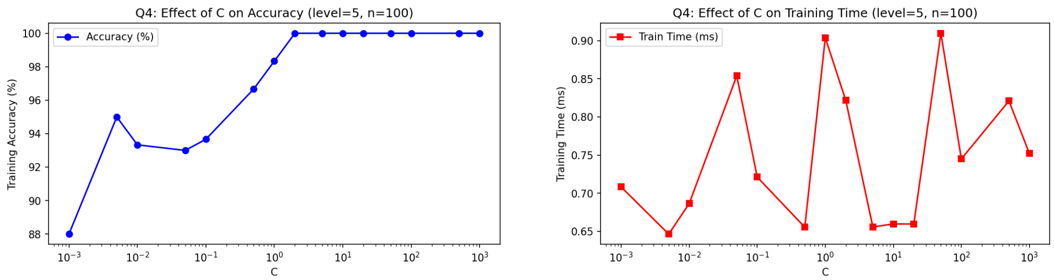

Figure 3: Effect of C on accuracy (left) and training time (right).

Table 4: Q4: Effect of `max_iter` on accuracy and training time

max_iter	Accuracy	Time (s)
1	92.00%	.0010
2	92.00%	.0008
3	93.67%	.0007
5	98.33%	.0006
7	98.33%	.0006
10	98.33%	.0008
15	98.33%	.0006
20	98.33%	.0006
50	98.33%	.0006
100	98.33%	.0006
500	98.33%	.0006
1000	98.33%	.0006
5000	98.33%	.0008
10000	98.33%	.0008

4.3 Effect of `tol` (Convergence Tolerance)

Analysis: For `tol` ≤ 0.05, accuracy remains at 98.33%. At `tol`=0.1, accuracy drops to 96%, and at `tol`≥ 5, the solver terminates so early it predicts all one class (33.33%). Training time is roughly constant since the problem is small ($n = 100$, $d = 2$).

5 Question 5: Effect of Sample Size (10 marks)

Settings: `level`=100 (high challenge), $C = 1$, `max_iter`=1000, `tol`=10^{-4}.

Analysis:

- **Accuracy:** At $n = 10$, 100% accuracy is achieved trivially (few points, easy to separate). As n grows, accuracy stabilizes around 97%–98% since the default $C = 1$ is insufficient for perfect separation at high challenge levels. More points expose the overlap between $\boldsymbol{\mu}_{\text{pos1}}$ and $\boldsymbol{\mu}_{\text{neg}}$, which share nearly identical x_1-coordinates at `level`=100.
- **Training time:** Time scales roughly linearly with n, from $\sim$0.7 ms at $n = 10$ to $\sim$1.85 ms at $n = 1000$. The liblinear solver has near-linear complexity in the number of training points.

References

- Scikit-learn LinearSVC documentation: `https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html`
- Course materials: CS771, IIT Kanpur, 2025–26.

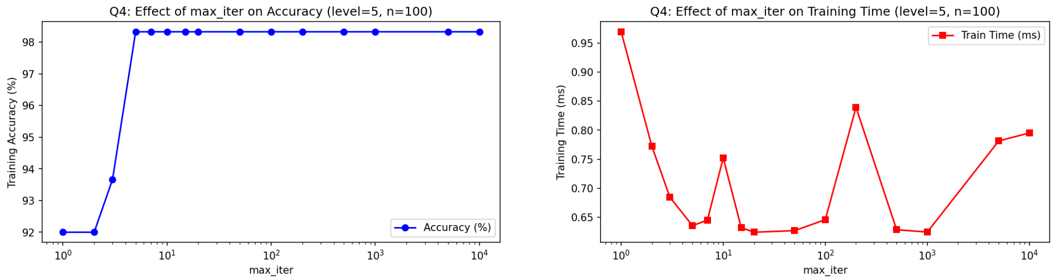

Figure 4: Effect of `max_iter` on accuracy (left) and training time (right).

Table 5: Q4: Effect of `tol` on accuracy and training time

tol	Accuracy	Time (s)
10^{-6}	98.33%	.0008
10^{-5}	98.33%	.0007
10^{-4}	98.33%	.0007
5×10^{-4}	98.33%	.0007
0.001	98.33%	.0007
0.005	98.33%	.0007
0.01	98.33%	.0007
0.05	98.33%	.0008
0.1	96.00%	.0006
0.5	92.00%	.0007
1.0	92.00%	.0007
5.0	33.33%	.0006
10.0	33.33%	.0008
50.0	33.33%	.0006

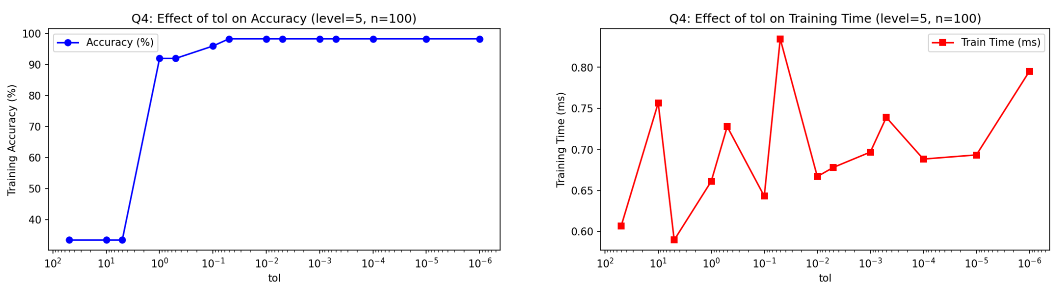

Figure 5: Effect of `tol` on accuracy (left) and training time (right).

Table 6: Q5: Accuracy and training time for varying sample sizes (`level`=100)

n	Accuracy	Time (ms)
10	100.00%	0.90
25	94.67%	0.79
50	97.33%	0.67
75	95.56%	0.74
100	97.33%	0.85
200	97.83%	0.97
500	97.13%	1.17
750	97.07%	1.84
1000	97.93%	1.85

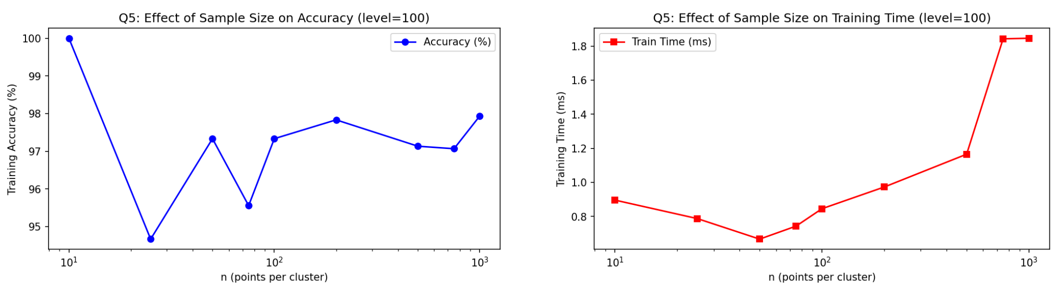

Figure 6: Effect of sample size n on accuracy (left) and training time (right) at `level`=100.